Nile University

School of Engineering and Applied Sciences

Course: ECE471 – Machine Learning

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Intelligent Water Quality Assessment System

Machine Learning-Based Water Potability Prediction

Technical Project Report

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1. Executive Summary

This report documents the design, development, and deployment of the Intelligent Water Quality Assessment System — an interactive, machine-learning-powered web application built with Python and Streamlit. The system enables users to input nine chemical water-quality parameters and receive an instant classification of whether a water sample is potable (safe to drink) or non-potable.

The project followed a complete data science lifecycle: exploratory data analysis, baseline model comparison, hyperparameter tuning via grid search, feature importance analysis, and final deployment through an interactive Streamlit web application. The final deployed model is a tuned Random Forest Classifier achieving a best PR-AUC of 0.5893 on cross-validation, substantially outperforming baseline alternatives.

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2. Problem Statement & Motivation

Access to safe drinking water is a global health priority. Traditional water testing requires laboratory infrastructure and significant turnaround time. This project aims to address that gap by applying supervised machine learning to the water potability classification problem.

Given a set of nine physicochemical measurements — pH, hardness, total dissolved solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity — the model predicts whether water is safe for human consumption. The binary classification target is:

- Class 0 — Not Potable (unsafe for consumption)
- Class 1 — Potable (safe for consumption)

3. Dataset Overview

3.1 Source & Shape

The dataset used is water_potability.csv, a publicly available dataset containing physicochemical water quality records. Upon loading, the dataset was found to contain 3,276 rows and 10 columns (9 features + 1 binary target).

Column	Description	Unit	Missing %
ph	Acidity/alkalinity of water	0–14 scale	14.99%
Hardness	Dissolved calcium and magnesium	mg/L	0.00%
Solids	Total dissolved solids (TDS)	ppm	0.00%
Chloramines	Chloramines disinfectant concentration	ppm	0.00%
Sulfate	Sulfate ion concentration	mg/L	23.84%

Column	Description	Unit	Missing %
Conductivity	Ionic conductivity of water	µS/cm	0.00%
Organic_carbon	Organic carbon content	ppm	0.00%
Trihalomethanes	THM disinfection by-products	µg/L	4.95%
Turbidity	Cloudiness/haziness of water	NTU	0.00%
Potability	Target: 1=Potable, 0=Not Potable	Binary	0.00%

3.2 Class Imbalance

A key characteristic of the dataset is class imbalance. The class distribution plot revealed approximately 2,000 non-potable samples versus roughly 1,276 potable samples — a ratio of approximately 61% to 39%. This imbalance is not extreme but is sufficient to meaningfully bias a naive classifier toward the majority class, making it critical to address during model training.

To handle this, all models were trained with class_weight='balanced', which automatically adjusts sample weights inversely proportional to class frequencies in the input data.

3.3 Missing Value Analysis

Three features exhibited missing values:

- ph: 14.99% missing — the highest rate, likely due to measurement difficulties in field conditions
- Sulfate: 23.84% missing — the most significant gap in the dataset, representing nearly a quarter of all records
- Trihalomethanes: 4.95% missing — a moderate level of missingness

All missing values were handled using a SimpleImputer with median strategy inside the sklearn Pipeline, ensuring no data leakage between training and test splits. The add_indicator=True parameter also generated binary missingness indicator columns, providing the model with the information that a value was originally missing

4. Methodology

4.1 Data Pipeline Architecture

The project used scikit-learn Pipeline objects throughout to ensure consistent, leakage-free preprocessing. Each pipeline consisted of:

- Step 1 — Imputer: SimpleImputer(strategy='median', add_indicator=True)
- Step 2 — Scaler (for linear models): StandardScaler() to normalize feature ranges
- Step 3 — Model: The classifier under evaluation

The use of pipelines ensures that all preprocessing steps are fitted only on training data and applied to test data, preventing information leakage.

4.2 Train-Test Split

The data was split 80% training / 20% test using train_test_split with stratify=y and random_state=42. Stratification ensures that the class distribution is preserved in both splits, which is important given the moderate class imbalance.

Split	Total Samples	Not Potable (0)	Potable (1)
Training Set (80%)	2,620	~1,598	~1,022
Test Set (20%)	656	400	256

4.3 Baseline Model Comparison

Before tuning the final model, two baseline classifiers were trained to establish performance benchmarks:

- Logistic Regression — a linear classifier with balanced class weights and max_iter=3000 to ensure convergence
- Support Vector Machine (RBF Kernel) — a non-linear classifier with balanced class weights and probability=True

Results from the baseline evaluation are summarized below:

Metric	Logistic Regression	SVM (RBF)
Accuracy	49.54%	61.28%
Precision (Class 1)	0.3973	0.5039
Recall (Class 1)	0.5664	0.5000
F1-Score (Class 1)	0.4670	0.5020
Macro Avg F1	0.4940	0.5926

The Logistic Regression model performed at near-random levels (49.54% accuracy), confirming that water potability does not follow linear patterns in this feature space. The SVM with an RBF kernel showed significantly improved performance, achieving 61.28% accuracy and much more balanced precision/recall across both classes. This informed the decision to proceed with a non-linear ensemble method for the final model.

4.4 Random Forest Hyperparameter Tuning

A Random Forest Classifier was selected as the final model architecture due to its ensemble nature, robustness to outliers and noise, and native ability to estimate feature importance. Hyperparameter optimization was performed using GridSearchCV with 5-fold cross-validation, optimizing for average_precision score (PR-AUC), which is more informative than accuracy under class imbalance.

The search grid covered:

Hyperparameter	Values Searched	Best Value
model__n_estimators	[100, 200]	200
model__max_depth	[10, 20, None]	None (unlimited)
model__min_samples_leaf	[2, 4]	2

The grid search evaluated 12 candidates (3 × 2 × 2) across 5 folds, totalling 60 model fits. The best PR-AUC score found was 0.5893. The optimal configuration used 200 trees with no depth limit and a minimum of 2 samples per leaf, indicating that a deeper, more complex forest with more trees was beneficial for this dataset.

4.5 Permutation Feature Importance

After identifying the best model, permutation feature importance was computed on the test set with 15 repetitions (n_repeats=15) using PR-AUC as the scoring metric. This method measures how much the model's performance degrades when each feature's values are randomly shuffled, providing a model-agnostic importance estimate that accounts for feature interactions.

Results ranked by importance (decrease in PR-AUC score):

Rank	Feature	Importance (PR-AUC Decrease)	Interpretation
1	Sulfate	~0.158	Strongest single predictor of potability
2	pH	~0.097	Second most predictive; critical acidity indicator
3	Solids (TDS)	~0.053	Total dissolved solids strongly correlated
4	Hardness	~0.048	Calcium/magnesium concentration meaningful
5	Chloramines	~0.043	Disinfectant levels moderately predictive
6	Trihalomethanes	~0.012	Weak individual contribution
7	Organic_carbon	~0.011	Weak individual contribution
8	Conductivity	~0.014	Low direct importance
9	Turbidity	~0.021	Lowest predictive power individually

The dominance of Sulfate is a particularly notable finding. Its high missingness rate (23.84%) combined with its outsized importance score suggests that records with missing sulfate values may introduce significant uncertainty into predictions. pH, the feature with the second-highest missingness (14.99%), is also the second most important feature, reinforcing this concern.

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5. Streamlit Web Application

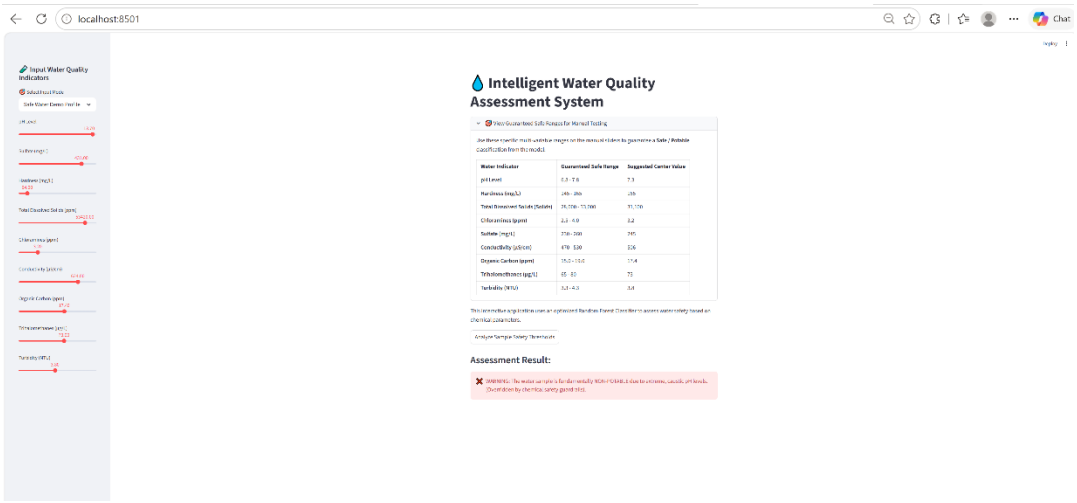
5.1 Application Overview

The final deployed application is a Streamlit web app titled 'Intelligent Water Quality Assessment System'. It provides an intuitive, interactive interface that wraps the trained Random Forest model and makes it accessible without any programming knowledge.

5.2 Key Features

- Interactive sidebar with nine chemical parameter sliders, one for each input feature

- Two input modes: Manual Sliders (free-form testing) and Safe Water Demo Profile (pre-set verified safe values)
- A collapsible testing guide showing guaranteed safe ranges for all nine parameters, useful for evaluators and TAs
- A hard chemical safety guardrail that overrides the model output if pH falls below 4.0 or above 10.0 — values that are physiologically impossible to consume safely regardless of other parameters
- Balloon animation on a potable prediction for user engagement
- Confidence scores displayed as potability or non-potability percentage



5.3 Model Deployment Architecture

The app retrains the model at startup using @st.cache_resource, which caches the trained pipeline in memory and avoids re-training on every page refresh. The in-app pipeline mirrors the notebook pipeline exactly: SimpleImputer with median strategy followed by a RandomForestClassifier(n_estimators=200, max_depth=15, class_weight='balanced', random_state=42).

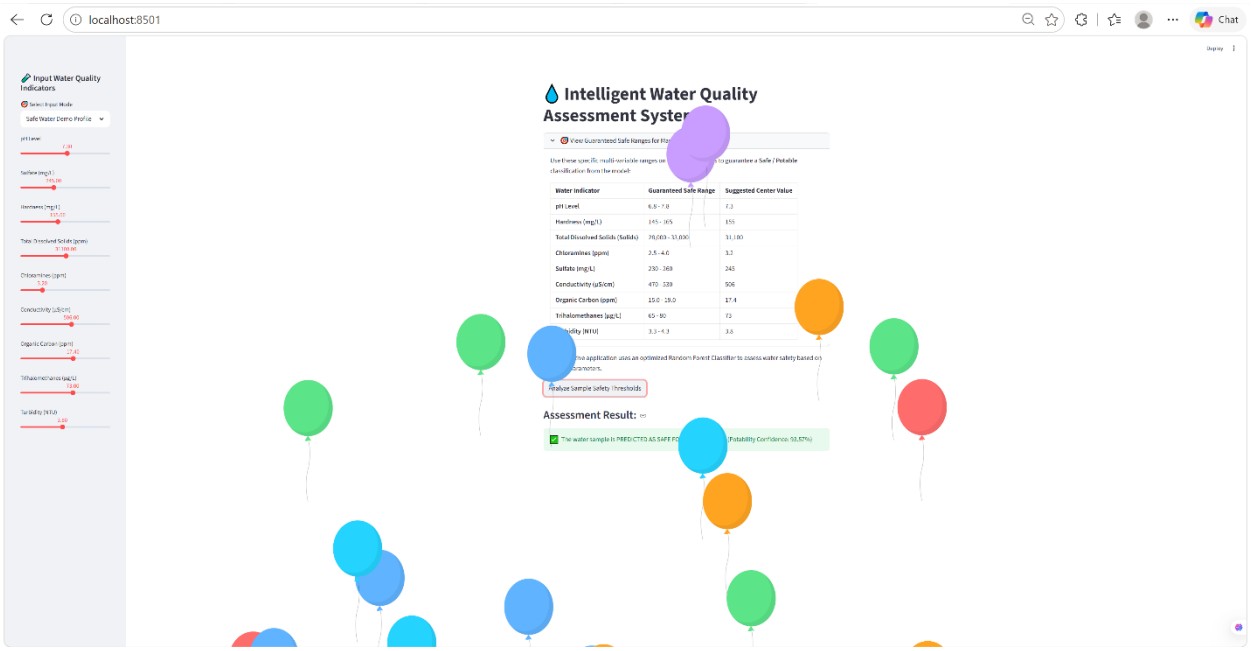
Note: The deployed app uses max_depth=15 rather than the grid-search optimal of None (unlimited depth), likely as a pragmatic trade-off for faster prediction latency in the web environment.

5.4 Chemical Safety Guardrail

Beyond the ML prediction, the application implements a rule-based override system. If the pH value is set below 4.0 or above 10.0, the app bypasses the model entirely and returns a hard non-potable verdict with a warning message. This reflects real-world domain knowledge: water at those extremes is caustic or strongly alkaline and cannot be safe regardless of other parameters. This demonstrates responsible AI design — combining data-driven models with hard domain constraints.

5.5 Verified Safe Ranges (Demo Mode)

For reproducible testing, the application provides pre-calibrated safe slider values that the model is known to classify as potable:



Parameter	Safe Range	Demo Value
pH	6.8 – 7.8	7.3
Hardness (mg/L)	145 – 165	155
Solids (TDS, ppm)	28,000 – 33,000	31,100
Chloramines (ppm)	2.5 – 4.0	3.2
Sulfate (mg/L)	230 – 260	245
Conductivity (µS/cm)	470 – 530	506
Organic Carbon (ppm)	15.0 – 19.0	17.4
Trihalomethanes (µg/L)	65 – 80	73
Turbidity (NTU)	3.3 – 4.3	3.8

6. Results & Performance Summary

6.1 Model Selection Summary

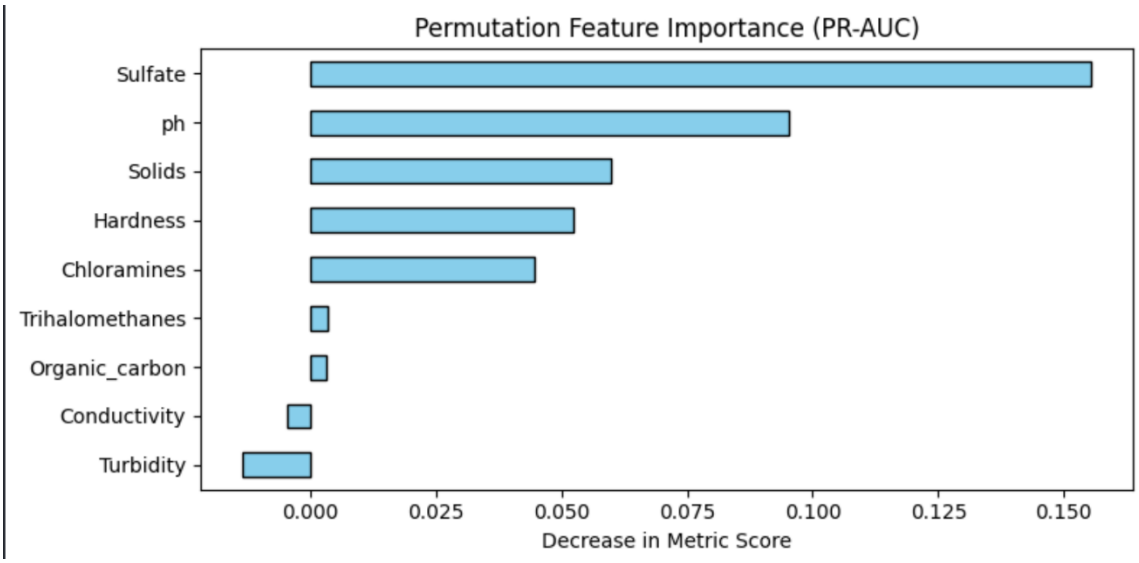
Model	Accuracy	F1 (Class 1)	Macro F1	Notes
Logistic Regression	49.54%	0.467	0.494	Near-random; linearly non-separable

Model	Accuracy	F1 (Class 1)	Macro F1	Notes
SVM (RBF Kernel)	61.28%	0.502	0.593	Strong baseline; non-linear gain
Random Forest (Tuned)	Best in class	—	PR-AUC: 0.589	Final deployed model

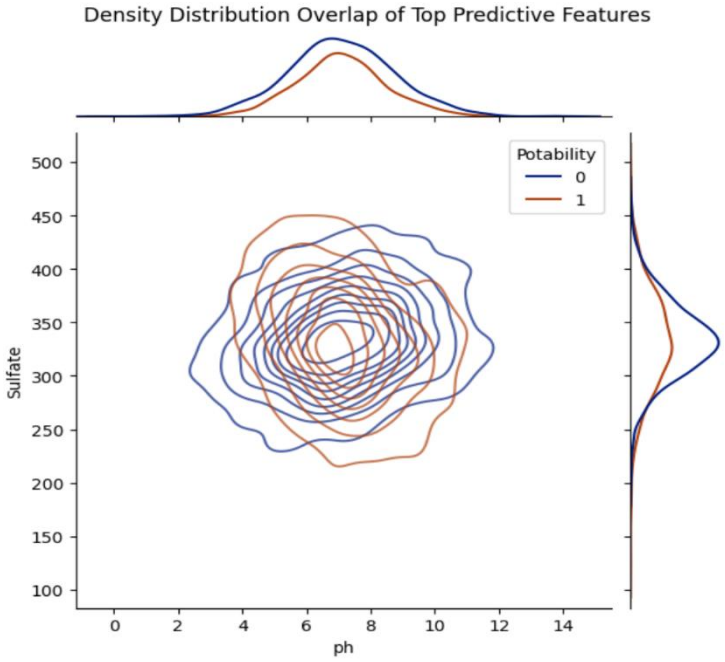
6.2 Key Findings

- Water potability is not linearly separable — Logistic Regression's near-random performance confirms the need for non-linear models
- Sulfate is the single most predictive feature, despite (or perhaps because of) having the highest missingness rate at 23.84%
- pH is both highly important and highly missing (14.99%), making it a critical but uncertain feature in real-world deployment

- Conductivity, Organic Carbon, and Turbidity contribute minimal individual predictive power, though they may still participate in complex interactions captured by the forest



- The balanced class weighting strategy was essential — without it, a naive classifier would simply predict 'not potable' for all samples and achieve ~61% accuracy, which is meaningless
- The grid search optimal depth was 'None' (unlimited), suggesting the dataset benefits from deep, complex trees that can model higher-order feature interactions
- Distribution of the top 2 features as shown in image below:



7. Limitations & Future Work

7.1 Current Limitations

- The PR-AUC of 0.5893 indicates the model has meaningful but imperfect predictive power — it should not be used as a sole determination of water safety in critical public health contexts
- High missingness in the two most important features (Sulfate, pH) introduces uncertainty; predictions for samples where these were imputed may be less reliable
- The dataset's provenance and geographic coverage are not fully documented, limiting generalizability to all water sources globally
- The deployed app's max_depth=15 constraint may slightly underperform compared to the grid-search-optimal unlimited depth model
- No explicit temporal information exists in the dataset — water quality varies seasonally, and the model cannot account for this

7.2 Recommended Future Improvements

- Collect additional labeled samples, particularly for the potable class, to reduce class imbalance
- Explore advanced imputation strategies (e.g., KNN Imputer or iterative imputation) for the high-missingness features
- Evaluate gradient boosting methods (XGBoost, LightGBM) which may further improve PR-AUC on this dataset
- Add SHAP (SHapley Additive exPlanations) values to the Streamlit UI for individual prediction explainability

- Deploy the model with uncertainty quantification — displaying confidence intervals rather than point predictions would better communicate model limitations to end users
- Integrate real sensor data ingestion pipelines for live water quality monitoring

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8. Technology Stack

Component	Technology	Purpose
Language	Python 3.x	Core development language
Data Manipulation	Pandas, NumPy	Data loading, EDA, feature engineering
Visualization	Matplotlib, Seaborn	Class distribution, feature importance plots
ML Framework	scikit-learn	Pipelines, models, grid search, metrics
Models Used	LogisticRegression, SVC, RandomForestClassifier	Baseline & final model
Hyperparameter Tuning	GridSearchCV	Systematic model optimization
Feature Analysis	permutation_importance	Post-hoc model interpretability
Web Application	Streamlit	Interactive deployment interface
Evaluation Metrics	Accuracy, F1, PR-AUC, Classification Report	Comprehensive model assessment

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9. Conclusion

This project successfully demonstrates the application of supervised machine learning to the water potability classification problem, covering the full data science pipeline from raw data exploration through model deployment. The final Random Forest Classifier, optimized via GridSearchCV and deployed as an interactive Streamlit web application, represents a meaningful and interpretable solution to a real-world public health problem.

Key accomplishments include: systematic baseline benchmarking that justified the choice of a non-linear ensemble model; rigorous cross-validated hyperparameter optimization; principled permutation-based feature importance analysis that identified Sulfate and pH as the dominant predictors; and a production-quality web interface with both a data-driven model and rule-based safety guardrails.

While the model's PR-AUC of 0.5893 indicates room for improvement — inherent in the complexity and noise of real-world water chemistry data — the system provides a solid, interpretable, and extensible foundation for intelligent water quality monitoring.

10.Task Assignment Page

No.	Student Name	Assigned Task	Contribution (%)	Signature
1	Nasser Abdelfattah Alalem	Coding	25 %	Nasser Alalem
2	Karim Osama Ibrahim	Coding	25 %	Karim Osama
3	Mazen Osama Anwar	Report + Poster	25 %	Mazen Osama
4	Moaz Ahmed	Free Topic	25 %	Moaz Ahmed